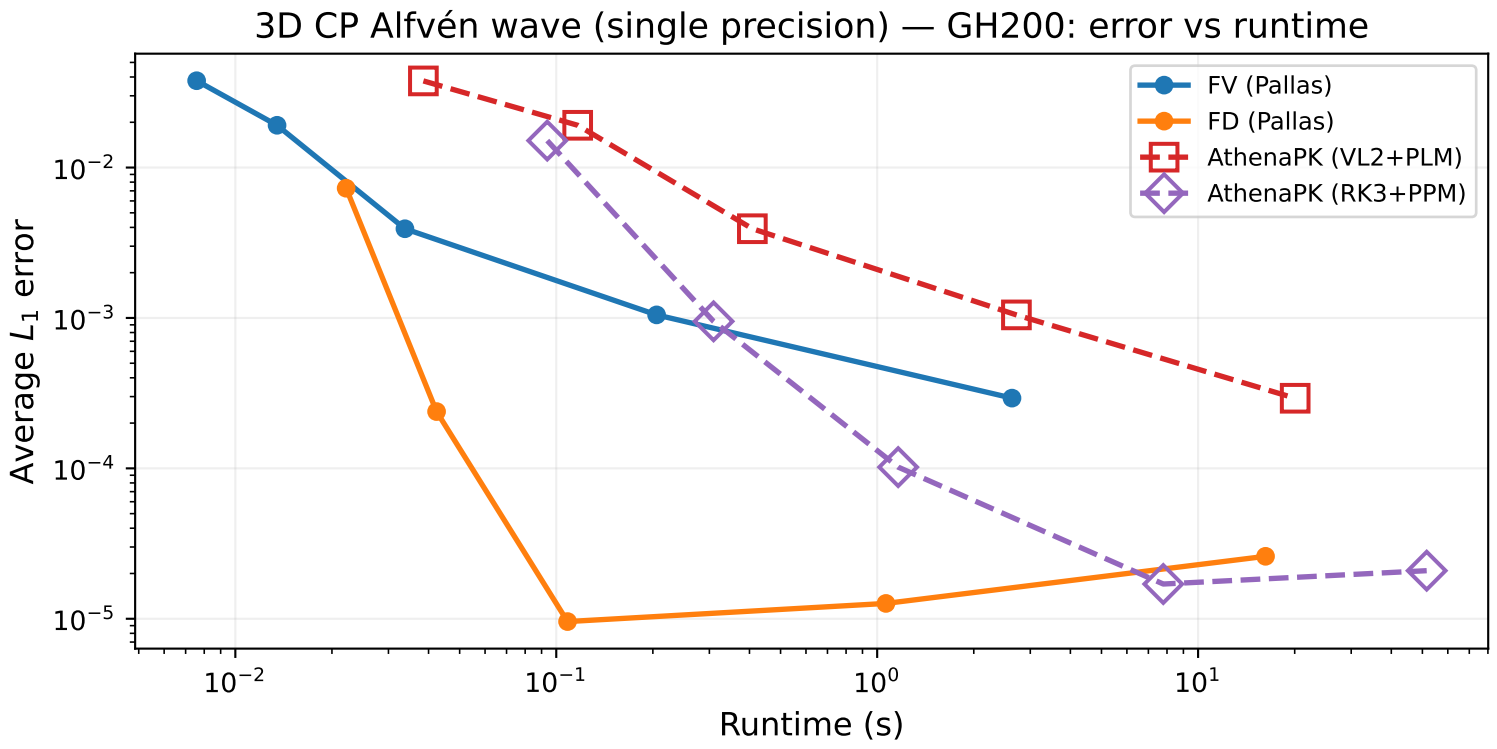
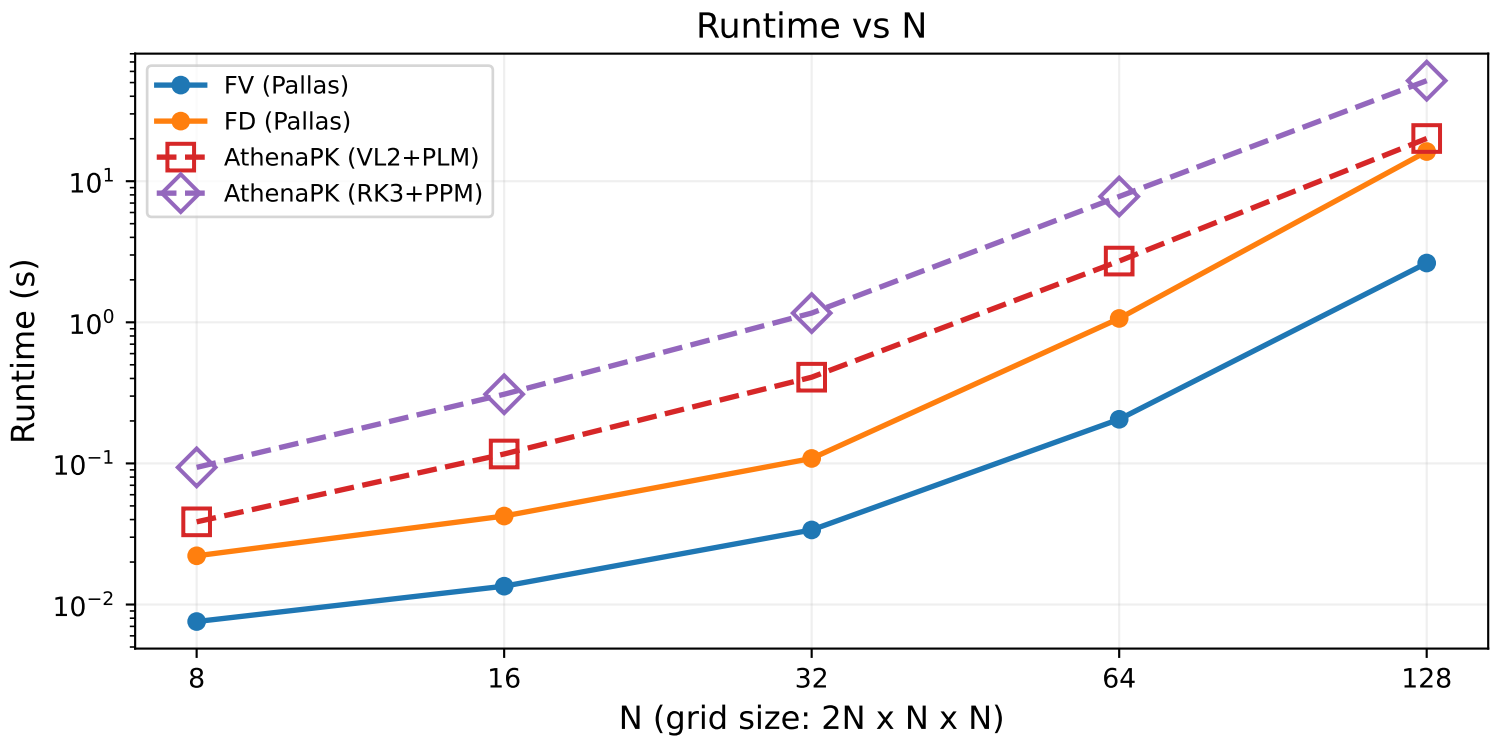
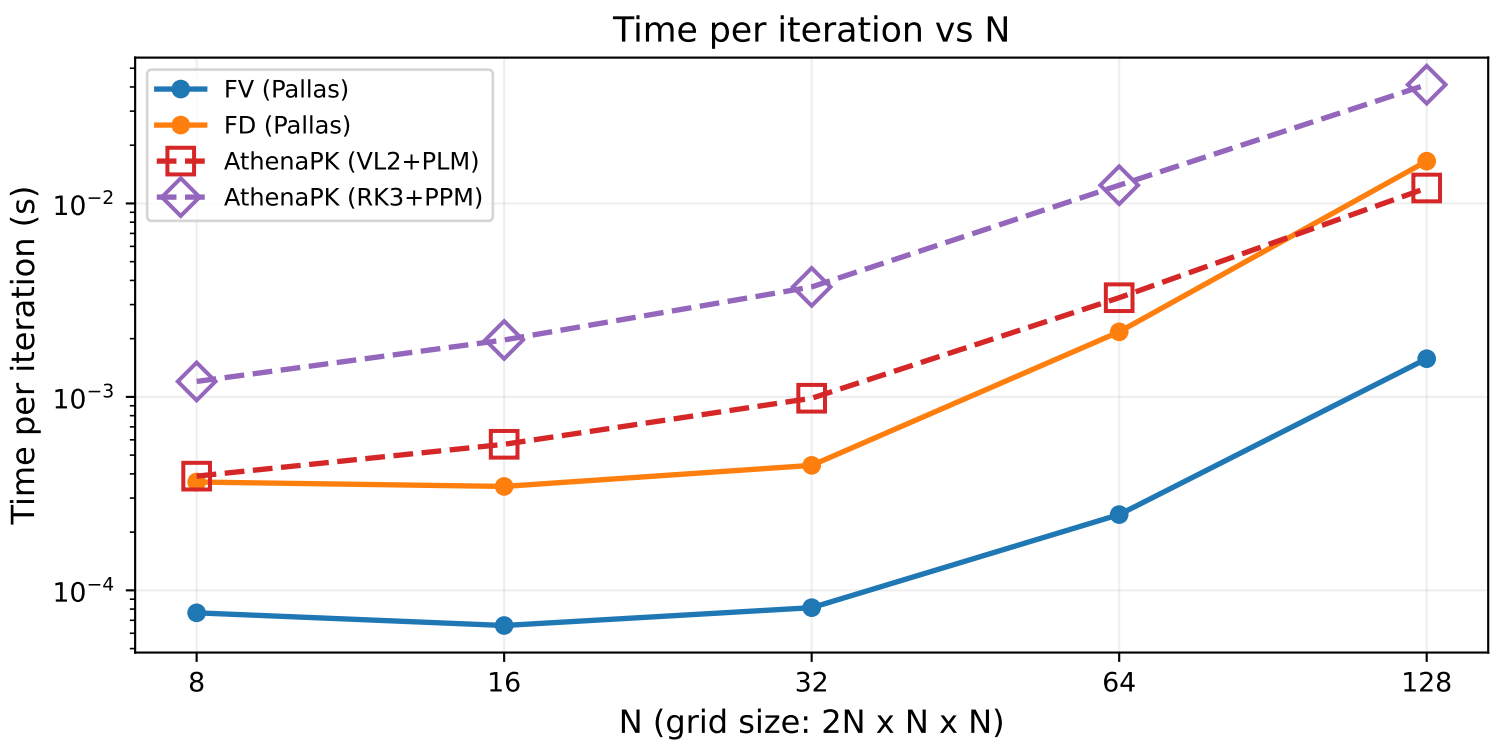

3D CP Alfvén wave (single precision) — GH200: error vs runtime
FV (Pallas)
FD (Pallas)
AthenaPK (VL2+PLM)
AthenaPK (RK3+PPM)
Average L_1 error
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-2}
10^{-1}
10^{0}
10^{1}
Runtime (s)


Runtime vs N
FV (Pallas)
FD (Pallas)
AthenaPK (VL2+PLM)
AthenaPK (RK3+PPM)
Runtime (s)
10^{1}
10^{0}
10^{-1}
10^{-2}
8
16
32
64
128
N (grid size: 2N x N x N)


Time per iteration vs N
FV (Pallas)
FD (Pallas)
AthenaPK (VL2+PLM)
AthenaPK (RK3+PPM)
Time per iteration (s)
10^{-2}
10^{-3}
10^{-4}
8
16
32
64
128
N (grid size: 2N x N x N)